Practice Questions

Candidate: Dunkan Gibson (dunkan.gibson)
Date: 10/24/2021 8:53:30 pm • Time spent: 03:00

Score: 75% Passing Score: 80%

Question 1: ✔ Correct

What is the frequency of 802.11a networking?

- 900 MHz
- 1.2 GHz
- 2.4 GHz
- 5.75 GHz

EXPLANATION

802.11a wireless operates in the 5.75 GHz range.

802.11b and 802.11g operate in the 2.4 GHz range.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 2: ✕ Incorrect

How many total channels are available for 802.11a wireless networks?

- 3
- ~~11~~
- 12
- 23
- 54

EXPLANATION

802.11a wireless uses the 5.75 GHz range which has a total of 23 channels.

802.11b and 802.11g use the 2.4 GHz range which has a total of 11 channels in the US.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 3: ✔ Correct

How many total channels are available for 802.11g wireless networks?

- 3
- 11
- 12
- 23
- 54

EXPLANATION

802.11b and 802.11g use the 2.4 GHz range which has a total of 11 channels in the US.

802.11a wireless uses the 5.75 GHz range which has a total of 23 channels.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 4: ✔ Correct

Your organization uses an 802.11b wireless network. Recently, other tenants installed the following equipment in your building:

- A wireless television distribution system running at 2.4 GHz
- A wireless phone system running at 5.8 GHz
- A wireless phone system running at 900 MHz
- An 802.11a wireless network running in the 5.725 - 5.850 GHz frequency range
- An 802.11j wireless network running in the 4.9 - 5.0 GHz frequency range

Since this equipment was installed, your wireless network has been experiencing significant interference. Which system is to blame?

- The 900 MHz wireless phone system
- ➡ The wireless TV system
- The 802.11a wireless network
- The 802.11j wireless network
- The 5.8 GHz wireless phone system

EXPLANATION

Because the 802.11b standard operates within the 2.4 GHz to 2.4835 GHz radio frequency range, the most likely culprit is the wireless TV distribution system.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 5: ✔ Correct

Which of the following are frequencies defined by 802.11 committees for wireless networking? (Select two.)

- [ ] 700 MHz
- [ ] 1.9 GHz
- [x] 2.4 GHz
- [x] 5.75 GHz
- [ ] 10 GHz

EXPLANATION

802.11 specifications for wireless include standards for operating in the 2.4 GHz range (802.11b, 802.11g, and 802.11n) and the 5.75 GHz range (802.11a and dual-band devices using 802.11n)

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 6: ✕ Incorrect

Which IEEE standard describes wireless communication?

- ( ) 802.11b
- ( ) 802.5
- ( ) 802.2
- (•) ~~802.3~~
- ( ) 802.7b

EXPLANATION

One IEEE standard for wireless is 802.11b.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 7: ✕ Incorrect

Which data transmission rate is defined by the IEEE 802.11b wireless standard?

- 2 Mbps
- 10 Mbps
- 11 Mbps
- ~~56 Mbps~~

EXPLANATION

The IEEE 802.11b standard defines wireless transmission rates up to 11 Mbps. Wireless network interface cards and wireless access points (also called wireless hubs or wireless routers) will automatically negotiate the best transmission speed up to 11 Mbps based on current network traffic load and the quality of the wireless connection between the client and access point. The wireless communications are affected by distance, dense physical obstructions, and other electromagnetic interference producing devices.

The IEEE 802.11a standard defines wireless transmission rates up to 2 Mbps. The IEEE 802.11g standard defines wireless transmission rates up to 56 Mbps. The IEEE 802.3 standard defines Ethernet 10baseT cable based transmissions of 10 Mbps.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 8: ✔ Correct

You are designing a wireless network for a client. Your client needs the network to support a data rate of at least 54 Mbps. In addition, the client already has a wireless telephone system installed that operates 2.4 GHz.

Which 802.11 standard will work best in this situation?

- 802.11g
- 802.11b
- 802.11c
- 802.11a
- 802.11d

EXPLANATION

802.11a is the best choice for this client. While both 802.11a and 802.11g each operate at 54 Mbps, 802.11g operates in the 2.4 GHz to 2.4835 GHz range; which will cause interference with the client's wireless phone system.

802.11a, on the other hand, operates in the 5.725 GHz to 5.850 GHz frequency range; which won't interfere with the phone system.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 9: ✔ Correct

You are designing an update to your client's wireless network. The existing wireless network uses 802.11b equipment; which your client complains runs too slowly. She wants to upgrade the network to run up to 600 Mbps.

Due to budget constraints, your client wants to upgrade only the wireless access points in the network this year. Next year, she will upgrade the wireless network boards in her users' workstations. She has also indicated that the system must continue to function during the transition period.

Which 802.11 standard will work best in this situation?

- 802.11a
- 802.11b
- 802.11d
- ➡ 802.11n
- 802.11c

EXPLANATION

802.11n is the best choice for this client and provides up to 600 Mbps. With 802.11n, you may have a single device that uses multiple radios, one that can operate at one frequency and another that can operate on a different frequency. Because of this, 802.11n usually allows for compatibility between all 802.11 standards, depending upon the specific implementation.

While 802.11g is compatible with 802.11b, it only provides up to 54 Mbps.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 10: ✔ Correct

Which IEEE wireless standards specify transmission speeds up to 54 Mbps? (Select two.)

- [ ] 802.1x
- [ ] Bluetooth
- [x] 802.11g
- [x] 802.11a
- [ ] 802.11b

EXPLANATION

Both the 802.11a and the 802.11g wireless standards specify maximum transmission speeds up to 54 Mbps.

Bluetooth is a wireless standard commonly used to connect peripheral devices and operates at 720 Kbps. The 802.11b wireless standard provides transmission speeds of 11 Mbps. 802.1x is a wireless security standard that provides an authentication framework for 802-based networks.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 11: ✔ Correct

You have been contacted by OsCorp to recommend a wireless Internet solution. The wireless strategy must support a frequency range of 5 GHz, and provide the highest possible transmission speeds.

Which of the following wireless solutions would you recommend?

- 802.11n
- 802.11a
- WEP
- 802.11b
- Bluetooth

EXPLANATION

Of the technologies listed, only the IEEE 802.11n wireless standard addresses the desired requirements.

Bluetooth operates at the 2.4GHz radio frequency range and is limited to theoretical speeds of 1 to 3 Mbps and transmission distances of 33 feet. The 802.11a wireless standard offers speeds of 54 Mbps but uses the 5 GHz frequency range. 802.11b uses the 2.4 GHz frequency range but supports only 11 Mbps transfer speeds.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 12: ✔ Correct

Which technologies are used by the 802.11ac standard to increase network bandwidth? (Select two.)

- [x] Eight MIMO radio streams
- [ ] Data compression
- [x] 160 MHz bonded channels
- [ ] Four MIMO radio streams
- [ ] 40 MHz bonded channels

EXPLANATION

To increase network bandwidth, the 802.11ac standard uses:

- Eight MIMO radio streams
- 160 MHz wide bonded channels

REFERENCES

LabSim for Network Pro, Section 10.2.

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